DTCommittee
Digital Target Committee
شركة هدف لتقنيه العلومات
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CHNICAL & SOFT SKILLS

acle Ecosystem (Fusion & EBS)

- Oracle Fusion AI Agent Studio
- Oracle Fusion REST APIs
- Oracle APEX
- Oracle SQL
- PL/SQL
- Standard PL/SQL APIs
- BI Publisher
- OTBI
- Oracle Cloud Infrastructure (OCI)
- Oracle Report Builder
- Oracle Workflow Builder

ftware Engineering & Languages

- Python
- Java
- C++
- JavaScript
- FastAP
- asyncpg
- Pydantic
- Streamlit
- REST APIs
- Data Structures
- Algorithms
- OOP
- Design Patterns

LLMs & NLP

- LLM Fine-Tuning
- LoRA
- SFT
- Knowledge Distillation
- Retrieval-Augmented Generation (RAG)
- Multi-Agent Orchestration
- vLLM
- LangChain
- Prompt Engineering
- Arabic NLP

Mohamed Ehab

Oracle Technical Consultant

PROFESSIONAL SUMMARY

Oracle Technical Consultant specializing in AI-augmented enterprise applications, with expertise across Oracle APEX, Oracle SQL, and PL/SQL. Proven track record designing natural-language-to-SQL reporting assistants and multi-agent AI solutions on Oracle Fusion AI Agent Studio, alongside custom/standard report modification and OAF-based module enhancement. Combines strong PL/SQL and REST/JSON integration capabilities with applied AI expertise in RAG, LLM fine-tuning, and multi-agent orchestration, delivering end-to-end solutions across Oracle Cloud, Azure, and AWS.

EDUCATION

- **B.Sc. Computer Science, AI Specialization**

Zagazig University Egypt — Faculty of Computers and Information Sciences | Egypt

- Graduation project selected among the top 5 projects in the faculty by the academic committee.

CERTIFICATIONS

- Agentic AI Certified Foundations Associate | Oracle 10/2026
- Oracle Fusion AI Agent Studio Certified Foundations Associate — Rel 1 | Oracle [Certificate] 05/2026
- Oracle APEX: Foundations | Coursera [Certificate] 07/2026
- Oracle Cloud Infrastructure AI Foundations | Coursera [Certificate] 08/2026
- OCI AI Foundations Associate (2025) | Oracle [Certificate] 05/2026
- AWS Certified AI Practitioner | Amazon Web Services (AWS) via Manara [Certificate] 2026
- Mastering Deep Learning | ScSkilled — Dr. Mostafa Saad Ibrahim [Certificate] 08/2024 – 03/2026
- Machine Learning Diploma | ScSkilled — Dr. Mostafa Saad Ibrahim [Certificate] 07/2023 – 07/2024
- CS224N: Natural Language Processing with Deep Learning | Stanford University (Self-Study)08/2025 – 01/2026

WORK EXPERIENCE

- **Oracle Technical Consultant** *05/2025 – Present*

- English NLP
- PyTorch
- scikit-learn
- Hugging Face Transformers
- LLaMA-Factory

mputer Vision

- YOLOv11
- YOLOv12
- ResNet
- Vision Transformer (ViT)
- Object Detection
- Instance Segmentation
- Edge Inference
- OCR
- OpenCV

tabases & Vector Stores

- Oracle SQL
- PostgreSQL
- Qdrant
- PgVector
- Alembic

rastructure & MLOps

- Oracle Cloud Infrastructure (OCI)
- Docker
- GitHub Actions
- CI/CD
- Microsoft Azure
- AWS
- Prometheus
- Grafana
- Weights & Biases
- Nginx
- Locust

owledge Transfer

- Organizes an industry-readiness study group for aspiring AI engineers, producing structured educational content and guiding members toward junior AI roles

oss-Functional Delivery

- Comfortable owning full project lifecycles — from requirements gathering through production rollout — across Oracle, AI, and cloud domains

lf-Directed Problem Solving

- Diagnoses and resolves complex

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- Deliver AI-augmented Oracle APEX and Oracle Fusion solutions for enterprise business functions, owning schema design, PL/SQL development, REST API integration, and security hardening across the full delivery lifecycle .
- Engineered a AI AGENT assistant For EBS using Oracle Apex with multi-layer security guardrails, achieving 94%+ accuracy, a 2s average response time, and zero unsafe query executions.
- Integration with ETA (Egyptian Tax Authority) to automate supplier invoice creation. , and an Oracle analytics dashboard.
- Architected a multi-agent business process assistant on Oracle Fusion AI Agent Studio, integrating Oracle Fusion REST APIs with a retrieval-grounded policy Q&A agent.
- Build custom and Modify standard reports and enhanced custom modules built on OAF pages

- **Junior AI Engineer — Part-Time, Remote** *02/2025 – 07/2025*

OZ-Tech | Egypt

- Trained and deployed YOLOv12 object detection and instance segmentation models achieving 90%+ Precision and 87% Recall across multiple production environments.
- Owned the complete ML pipeline end-to-end: data collection, annotation, augmentation, model training, evaluation, and production deployment on real-world industrial datasets.
- Integrated deployed models with Arduino-based edge devices, enabling real-time on-device inference at sub-100ms latency in resource-constrained environments.
- Collaborated with cross-functional teams in an agile environment to deliver operational AI systems on schedule, supporting internal training programs in detection and segmentation tasks.

- **AI Engineer Intern** *02/2025 – 05/2025*

RMG | Egypt Office Saudi Arabia

- Developed and shipped computer vision models for object detection and segmentation, driving projects from research prototype to live production delivery.
- Accelerated model iteration cycles by implementing structured evaluation pipelines, reducing review time and improving deployment readiness for production handoff.
- Collaborated with cross-functional AI teams to integrate computer vision models into operational workflows across the full project lifecycle.

PROJECTS

- **AI Invoice Automation Agent** *2026*

Infrarose

- Designed and built an AI workflow agent that automates supplier invoice

integration issues independently (e.g., Oracle HCM REST API troubleshooting, multi-agent orchestration design) with minimal guidance

intake for Oracle Fusion Cloud Payables: a user uploads a PDF invoice in a chat window, and the invoice is created and validated in Fusion in about one minute, with no manual data entry.

- Implemented PDF-to-JSON extraction with reconciliation checks between line amounts and the header total, and separate processing paths for PO and non-PO invoices.
- Integrated Fusion REST APIs for supplier, supplier site, PO header and PO line lookups, and for invoice creation and the validateInvoice action; managed agent state with persistent variables.
- Developed line matching by description and unit price instead of line order, and automated GL account classification for non-PO lines using a mapping table.
- Added human-in-the-loop checkpoints only for ambiguous supplier sites and final approval, keeping accountants in control of judgment calls.

- **AI-Powered NL-to-SQL Assistant** *2026*

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- Built an Oracle APEX tool letting non-technical HR users query the database in plain English; translated natural language into safe, read-only Oracle SQL via the Cohere Chat API (command-a-03-2025) and auto-executed it through DBMS SQL.
- Constrained generation to a single explicitly defined HR schema, rejecting any request referencing unknown tables or columns, and rendered results as a live formatted HTML report inline in APEX.
- Implemented defense-in-depth security: prompt-level restriction to SELECT/WITH statements, prompt-injection guarding, and a server-side guard validating every generated statement's keyword and regex-scanning for DML/DDL/DCL before execution; all output escaped via APEX ESCAPE.HTML.
- Reached 94%+ accuracy on schema-valid SQL generation, a 2s average response time, and zero unsafe query executions reaching the database.

- **Invoice Automation Portal** *2026*

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- Architected a full REST-to-database import pipeline pulling live invoices from an external e-invoicing API (Egyptian ETA-style) via Oracle APEX Web Source Modules and REST Data Sources, with credential-store authentication and duplicate-import protection.
- Parsed invoice header and line-item JSON with APEX JSON into normalized Oracle tables, and built editable Interactive Grids with full CRUD and lost-update protection for vendor/finance review.
- Built an AJAX-driven, one-click invoice submission workflow (custom JavaScript dynamic actions triggering APEX INIT WF) and a hand-built, print-ready HTML/CSS invoice rendering engine.
- Designed an executive analytics dashboard with Oracle JET charts (trends, top vendors, top invoices, distribution by vendor) and dual authentication (native APEX accounts + custom EBS SSO).

- **Oracle E-Business Suite (EBS) Project** *2026*

Al-Babtain Power & Telecom

  - Delivered custom and standard reports modification for Al-Babtain Power & Telecom on Oracle E-Business Suite (EBS).
  - Updated and enhanced custom modules built on OAF (Oracle Application Framework) pages, ensuring alignment with client-specific business requirements.

- **Oracle E-Business Suite (EBS) Project** *2026*

Kasrawy Automotive Group

  - Delivered custom and standard reports modification for Kasrawy Automotive Group on Oracle E-Business Suite (EBS).
  - Updated and enhanced custom modules built on OAF (Oracle Application Framework) pages, ensuring alignment with client-specific business requirements.

- **Multi-Agent HR Absence Assistant** *2026*

Infrarose

  - Designed a two-agent orchestrated architecture on Oracle Fusion AI Agent Studio: an absence-management agent handling creation/retrieval against Oracle Fusion HCM REST APIs, and a policy RAG agent answering leave-policy questions.
  - Built pre-validation logic checking an employee's existing absence records before submission to prevent overlapping or duplicate entries (resolving conflict error ANC-3405005).
  - Delivered a coordinating orchestration layer routing each query to the correct agent, presenting a single unified conversational interface, and evolved the design to a supervisor/worker pattern for a manager demo.

- **Bilingual RAG System (Arabic/English)** *01/2026 – Present*

GitHub

  - Architected a production-grade RAG API supporting Arabic and English with 902ms average end-to-end latency via fully async FastAPI and asyncpg, with a provider-agnostic LLM interface supporting OpenAI GPT and Cohere Command-R+.
  - Designed a dual vector database layer (Qdrant + PgVector HNSW) using a factory pattern, enabling zero-code provider switching through environment configuration alone.
  - Built a parallelized OCR pipeline (ThreadPoolExecutor + Mistral OCR + Gemini Vision) to process scanned PDFs and mixed Arabic/English documents at scale.
  - Deployed to Microsoft Azure with CI/CD via GitHub Actions; instrumented 8 Docker Compose services behind Nginx with Prometheus and Grafana for full production observability.
  - Managed PostgreSQL schema lifecycle with Alembic migrations across three relational tables with JSONB metadata and indexed foreign keys, ensuring efficient chunk retrieval at scale.

- **ArabicLLM — Arabic NLP Fine-Tuning** *02/2026 – Present*

GitHub

  - Fine-tuned Qwen2.5-1.5B on Arabic NLP tasks (NER, summarization, keyword extraction) using LoRA rank 64 via LLaMA-Factory (SFT), matching GPT-4o-level structured output quality at zero per-request inference cost through full self-hosting.
  - Built a knowledge distillation pipeline using GPT-4o as teacher model to generate high-quality structured training data from raw Arabic news stories, validated with Pydantic v2 schemas.
  - Deployed a production vLLM 0.6.6 inference server with LoRA adapter hot-loading on an NVIDIA T4 GPU, sustaining 20 concurrent users at 14s average latency at 4096-token context, validated via Locust load testing.
  - Built a live Streamlit inference UI with real-time token streaming over an OpenAI-compatible API; tracked 3 training epochs at learning rate 2e-4 with Weights & Biases (W&B).